

IslandCare Medical Centre Practice SBA

Full School-Based Assessment project brief

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: IslandCare Medical Centre

Sector: Medicine

Description of Project

IslandCare Medical Centre provides general consultations, wellness checks and selected diagnostic services. Staff currently keep separate appointment and payment records. The centre wants a small information system to manage patients, emergency contacts, service types, appointments and outstanding balances.

Create a connected project that improves patient record keeping, analyses service charges, produces appointment communication and automates the calculation of charges and discounts.

How to use this practice SBA

- Read the whole project before starting any software task.
- Complete the sections in order so information remains consistent from one file to the next.
- The wording and data are original SPARK practice material. Your teacher's current assignment takes priority.
- The practice task headings follow the style of common CSEC school assignments. Current CXC limits should still be followed.

Contents

Section	Practice weighting
Database Management	20 practice marks
Spreadsheet	20 practice marks
Word Processing	20 practice marks
Web Page Design	10 practice marks
Problem-Solving and Programming	15 practice marks

Database Management - 20 practice marks

Task A - Tables

1. Create a database named IslandCareDB_YourFullName.accdb.
2. Create the Patient, Service and Appointment tables using suitable field names, data types and primary keys.
3. Enter realistic records and apply the validation rules required by the project.
4. Create the required relationships and enforce the correct links between the tables.

Task B - Queries and Form

1. Create a query called Daily Appointments. Show appointments for a date entered by the user. Include patient name, service and status.
2. Create a query called Outstanding Accounts. Join Patient, Service and Appointment and calculate the balance.
3. Create Patient Information Form with Appointment subform and clear patient-contact headings.

Task C - Report

1. Create Service Income Report grouped by ServiceName with visit count, total fees and average fee.

Spreadsheet - 20 practice marks

Task A - Calculations and Functions

1. Create MedicalAccounts_YourFullName.xlsx with worksheets named Accounts and Dashboard. Enter or import the project data into Accounts and format it as a clear data table.
2. Create the main calculations using efficient formulas. The SPARK reference uses formulas such as =F2-G2, =IF(D2>=65,E2*5%,0) and =E2-F2-G2+\$M\$2.
3. Use at least three suitable functions. Functions used in the SPARK reference include VLOOKUP, COUNTIF, AVERAGE, IF, SUM.

Task B - Analysis and Presentation

1. Sort and filter the data to answer this requirement: Show visits with Balance greater than 0 and Status not equal to Closed.
2. Create a summary showing Summary of total charges by ServiceName. Then create a suitable chart.
3. Create a linked total or key result on the Dashboard worksheet so that it updates when the detailed sheet changes.

Word Processing - 20 practice marks

Task A - Fillable Form

1. Create a document named Patient Information Form_YourFullName.docx. The form must collect the information needed for this project and include clear instructions.
2. Use suitable form controls. The SPARK reference uses: Patient name text box; Appointment date picker; Service drop-down list; Consent check box.

Task B - Mail Merge

1. Create a professional main document called Appointment and Balance Notice. Use the project data as the mail-merge source.
2. Insert the following merge fields where they make sense: PatientName, AppointmentDate, ServiceName, Balance.
3. Complete the final merged output and include one relevant table, chart or other imported item from the connected SBA work where appropriate.

Web Page Design - 10 practice marks

Web Page Design

1. Create one web page for IslandCare Medical Centre. The page must be suitable for the intended audience and include a project logo.
2. Organise the page into these content sections: Welcome, Services, Before Your Visit, Opening Hours, Contact.
3. Add at least two suitable hyperlink types and test them.
4. Add at least one relevant graphic and make sure the page remains readable at different screen sizes.
5. Save, test and prepare the final web-page evidence.

Problem-Solving and Programming - 15 practice marks

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the outstanding balance for each visit and the combined outstanding balance for all visits processed.
2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as MedicalBalances_YourFullName.pas.
2. Test the program and capture evidence that the expected and actual results agree.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

Submission checklist

- IslandCareDB_YourFullName.accdb
- MedicalAccounts_YourFullName.xlsx
- Patient Information Form_YourFullName.docx
- Appointment and Balance Notice_YourFullName.docx
- One completed web page with its supporting files
- MedicalBalances_YourFullName.pas
- ProgramDocumentation_YourFullName.pdf

Final reminder

Keep the same names, codes, dates, fees and rules across your database, spreadsheet, word-processing, web and programming work. A strong SBA reads like one connected project.